Birla Institute of Technology & Science, Pilani

Work Integrated Learning Programmes Division

Course: AIML-CZG-567 (AI & ML Techniques for Cyber Security)
Assignment-02: Intrusion Detection using Machine Learning Models

A. Problem Statement

Build a network intrusion detector, a predictive model capable of distinguishing between **bad (Attacks)** and **good (Normal)** connections.

B. Dataset to Use

KDD Cup 1999 dataset (https://www.kaggle.com/datasets/kavl31/kdd-cup-1999-data)

C. Process Steps

Steps involved to create the text summary

- Data pre-processing
- Data correlation
- Feature selection
- Modelling
    - Naïve Bayes algorithm
    - Decision Tree algorithm
    - Random Forest algorithm
    - SVM algorithm
    - Logistic Regression algorithm
    - Gradient Boosting algorithm
- Validation & Comparison among different algorithms

D. Prerequisites

- Python 3
- NLTK Toolkit
- IDE or Text Editor

E. Submission Instructions

A PDF document has to be uploaded on Taxilla under 'Assignment' covering following:

- Overall process description & solution approach
- Tool used and reasons to use this specific tool
- Source code in VM
- Source code snippets
- Final output results and analysis of results

Note: Each document page should have student's BITS Id.

F. References

https://www.geeksforgeeks.org/intrusion-detection-system-using-machine-learning-algorithms/

G. Evaluation Criteria

The assignment is for 10 marks. Following evaluation scheme will be used to grade the assignments:

S.No.	Evaluation Task	Marks
1	Overall solution design and process architecture	2
2	Tool used and reasons to use this specific tool	1
3	Final output results and analysis of results	2
4	Document quality (structure, detailing, presentation etc)	2
5	Additional creativity/innovation by the student	3